

Data Driven Economics

Text Mining in Economics

Lecture 4

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¹The opinions expressed are those of the author and do not reflect the views of the Bank of Italy, the ECB or the Eurosystem.

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Roadmap

1. Latent Dirichlet Allocation (LDA) for analyzing Text corpora.
2. Thorsrud (2018): how LDA can be used in economics
3. D'Amuri & Marcucci (2017): Google Trends for nowcasting & forecasting
4. Angelico, Marcucci, Miccoli & Quarta (2019): how LDA can be used in economics with social network data
5. Conclusion

Text Mining for Economic Applications

Steps and algorithms in Text mining applications

1. **Analytical Pre-Processing** ✓
 - 1.1 Remove punctuation and numbers
 - 1.2 Tokenization
 - 1.3 Case folding
 - 1.4 Stop words removal
 - 1.5 Stemming or Lemmatization
 - 1.6 Part-of-Speech Tagging
 - 1.7 Term-Document Matrix (tdm) and Inverse Document Frequency (idf)
2. **Dimensionality reduction techniques** ✓
 - 2.1 Boolean Techniques ⇐
 - 2.2 Dictionary Techniques ⇐
 - 2.3 Weighting Words ⇐
 - 2.4 Vector Space Models ⇐
 - 2.5 Latent Semantic Analysis (LSA) ⇐
 - 2.6 Latent Dirichlet Allocation (LDA) ⇐
 - 2.7 Descending Hierarchical Classification ⇐
 - 2.8 Supervised Machine Learning ⇐

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Latent Dirichlet Allocation (LDA)

Some Intuition

- ▶ Imagine you have three topics:

Topic 1 $\Rightarrow$ politics

Topic 2 $\Rightarrow$ fashion

Topic 3 $\Rightarrow$ sports

- ▶ Assume you have a recipe, i.e. 50% of Topic 1, 30% of Topic 2 and 20% of Topic 3
- ▶ Each topic consists of a set of words times their probabilities $word_j \times Pr_{j,k}$
- ▶ You take this recipe and generate a document based on this model rules
- ▶ You obtain a document with words from each of the three topics with different frequencies.
- ▶ But, in reality it is the reverse!
- ▶ You have to learn a topic model from a set of words appearing in a set of documents



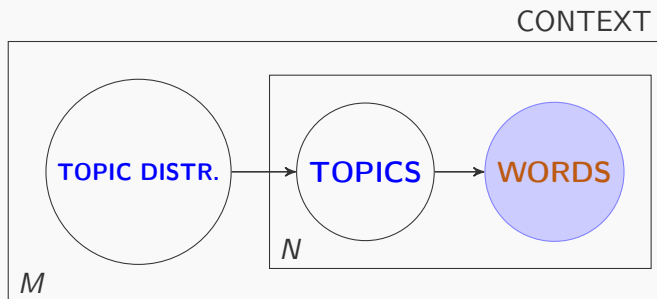
Latent Dirichlet Allocation (LDA)

Some Intuition

- ▶ Important Choices:
 - ▶ **How many topics?**
 - ▶ How are the topics assigned to a document?
- ▶ LDA is a BOW (Bag-of-Word) model so there is no syntax rule!
- ▶ LDA Model uses a real document using all words assuming they are related and tries to learn a model that would explain how such document would have been generated from the beginning if we followed that scenario

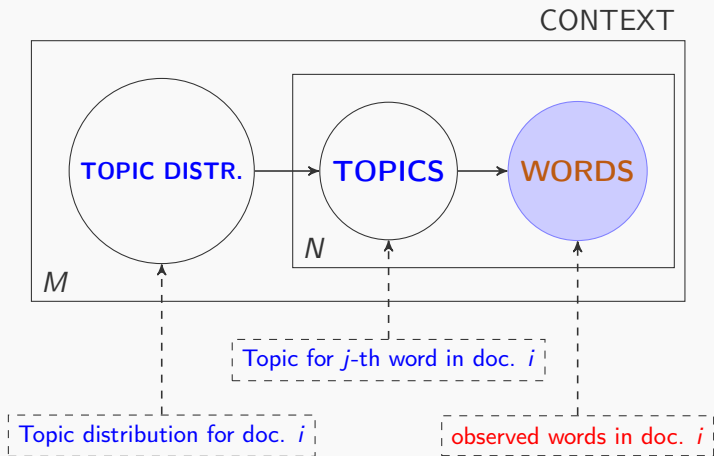
Latent Dirichlet Allocation (LDA)

- N words for M documents



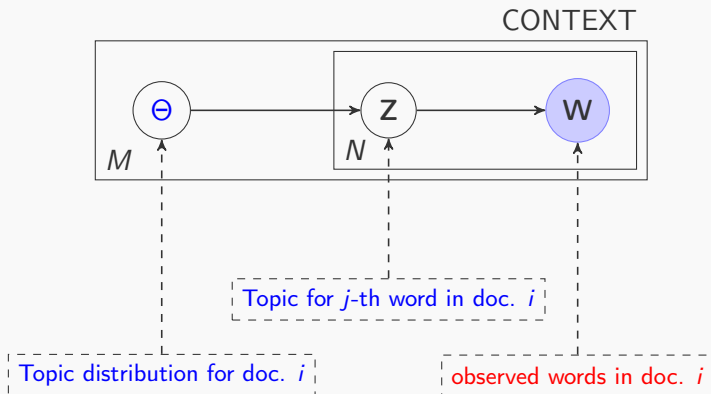
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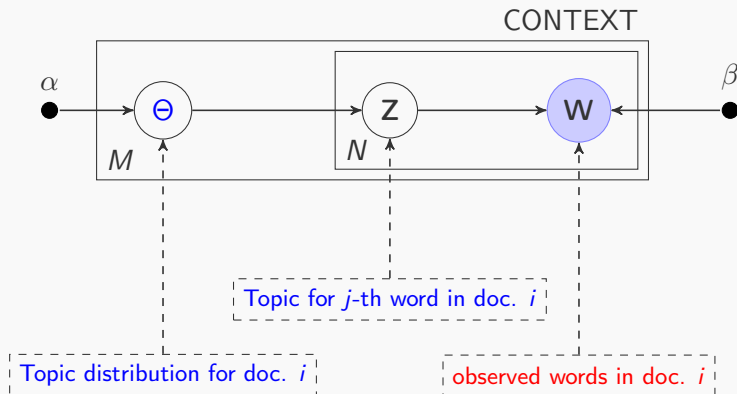
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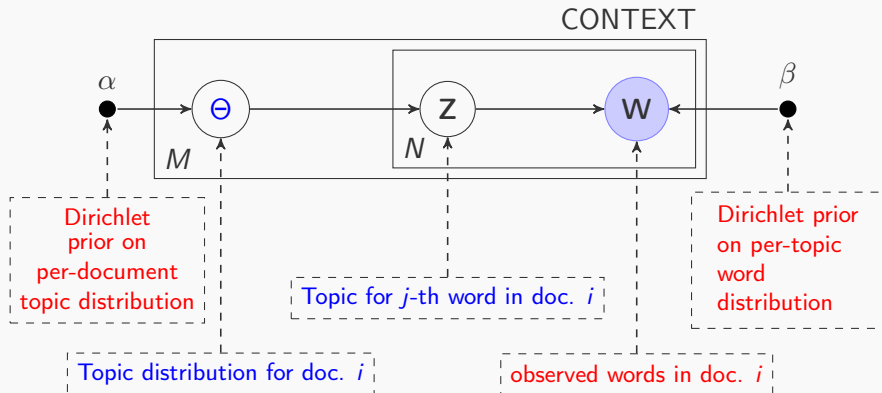
Latent Dirichlet Allocation (LDA)

- N words for M documents



Latent Dirichlet Allocation (LDA)

- N words for M documents



Latent Dirichlet Allocation (LDA)

Some Intuition

- ▶ Extra rules:
 - ▶ **Hyperparameter α**
 - ▶ **Hyperparameter β**
- ▶ α controls the **per document-topic distribution**
- ▶ β controls the **per topic-word distribution**
- ▶ A **high α** means that every document is likely to contain a mixture of most of the topics and not just any single topic
- ▶ A **low α** means that a document is more likely to be represented by just a few of the topics.
- ▶ A **high β** means that each topic is likely to contain any mixture of most of the words and not just any single words specifically.
- ▶ A **low β** means that a topic may contain a mixture of just a few of the words.

Latent Dirichlet Allocation (LDA)

Some Intuition

- ▶ In a nutshell:
- ▶ A **high** α will make document to appear more similar to each others.
- ▶ A **high** β will make the topics to appear more similar to each others.
- ▶ The final result:
- ▶ We get a file that contains all the topics made of all the words with their probabilities to belong to that topic.
- ▶ How many topics?
- ▶ It depends on the application domain.

Latent Dirichlet Allocation (LDA)

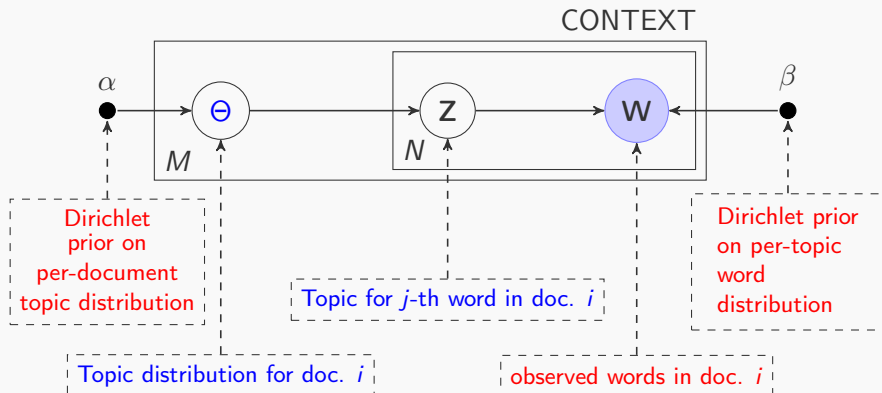
Some Intuition

- ▶ Think of LDA as a little robot guy head which has an understanding of words, topics and context and combines them all together.
- ▶ A document is a probability distribution over topics
- ▶ A topic is a probability distribution over words
- ▶ LDA model takes text and reads the BOW
- ▶ Remember that the text corpus and pre-processing are the more critical steps.

Latent Dirichlet Allocation (LDA)

- N words for M documents

$$P(W, Z, \theta, \varphi, \alpha, \beta) = \prod_{i=1}^K P(\varphi_i; \beta) \cdot \prod_{j=1}^M P(\theta_j; \alpha) \cdot \prod_{t=1}^N P(z_{j,t} | \theta_j) \cdot P(w_{j,t} | \varphi, z_{j,t})$$



Thorsrud (JBES, 2018)

Words are the New Numbers: A Newsy Coincident Index of the Business Cycle

By Leif Thorsrud

Journal of Business & Economic Statistics, 2018

What is the state of the economy today?

- ▶ The answer is important because...
 - ▶ Policy makers need to know the state of the economy in real time to devise appropriate policy responses
 - ▶ Forecasters would benefit from conditioning on a timely information set
- ▶ ...but, it is a difficult question
 - ▶ In real time, our best known measure of economic activity, GDP growth, is not observed:
 - ▶ It is registered on a quarterly frequency
 - ▶ It is published with a considerable lag

Traditional approaches

Use:

- ▶ National Account statistics and high-frequency (financial) data from structural databases
- ▶ Mixed-frequency time-series models (and model combination)

Result in:

- ▶ Coincident indexes: Stock and Watson [1988], Mariano and Murasawa [2003], Aruoba et al. [2009], Marcellino et al. [2013]
- ▶ Nowcasting: Giannone et al. [2008], Breitung and Schumacher [2008], Angelini et al. [2011], Kuzin et al. [2011]

Problems and opportunities

Problems:

- ▶ Few high-frequency variables reflecting the broader economy
- ▶ Instabilities across time (Stock and Watson [2003])

Opportunities:

- ▶ New technologies allow for alternative models and data sources, like (newspaper) text

Text as data?

1. Potentially available at a high frequency
2. Potentially reflecting the broader economy
3. A number is a fact, but the media in which it is presented/discussed/opinionated adds to the information
4. Rational (in)attention theory: Media an important channel for information diffusion

But, putting text into models is somewhat new (to economists) - until recently...

We show

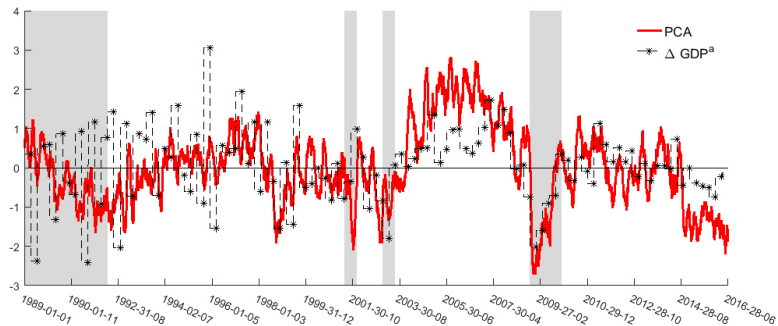
When textual news is appropriately classified using a topic model - it adds valuable information for understanding economic fluctuations

In particular, it can be used to:

1. Track the business cycle very well
2. Nowcast GDP growth

A short (in-sample) preview

Simple PCA decomposition of topics (unsupervised) and GDP growth



Key input:



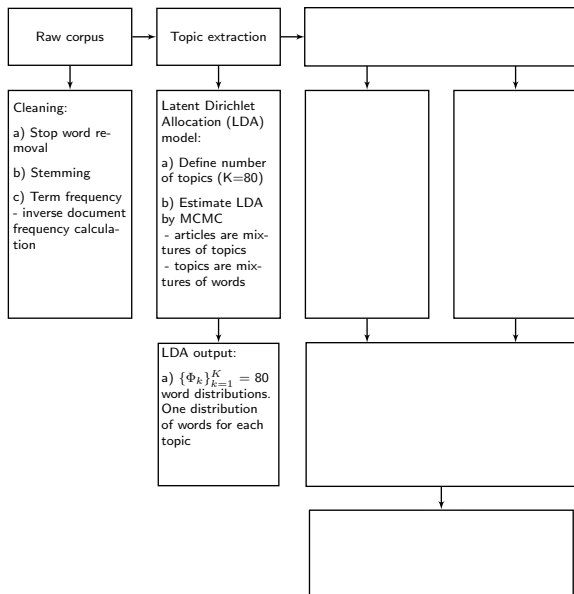
Use a primary source for new information directly - the newspaper!

Key hypothesis:

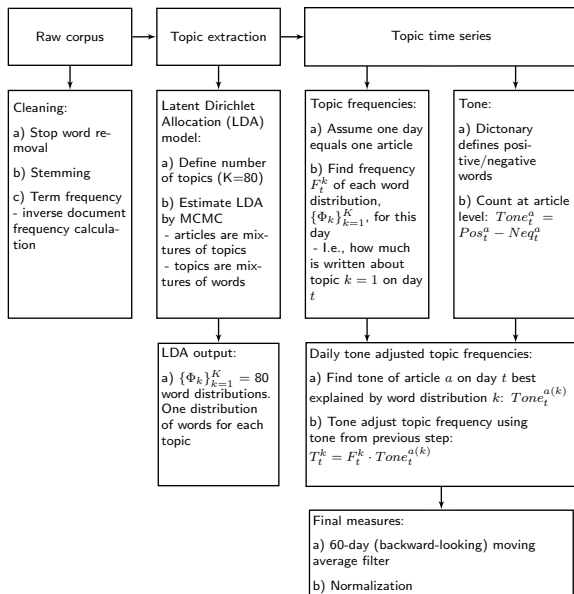
The more a newspaper writes about a topic the more likely it is that this topic reflects something of importance for the economy's current and future needs and developments

Two-step approach

1. Decompose the newspaper corpus into tone adjusted topic frequency time series
 - ▶ Cleaning the corpus
 - ▶ Finding the topics
 - ▶ Constructing time series
2. Develop a time-varying mixed frequency factor model to:
 - ▶ Estimate a daily news-based coincident index
 - ▶ Nowcast

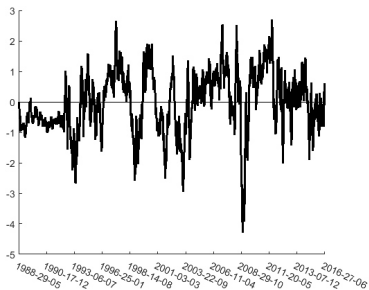


We call this the *Funding* topic

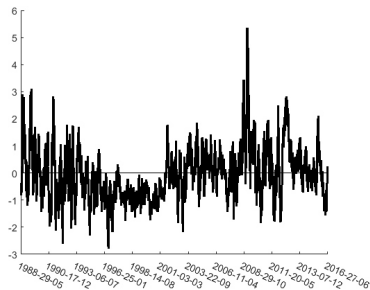


Topic time series

Topic: Monetary Policy (72)



Topic: Funding (42)



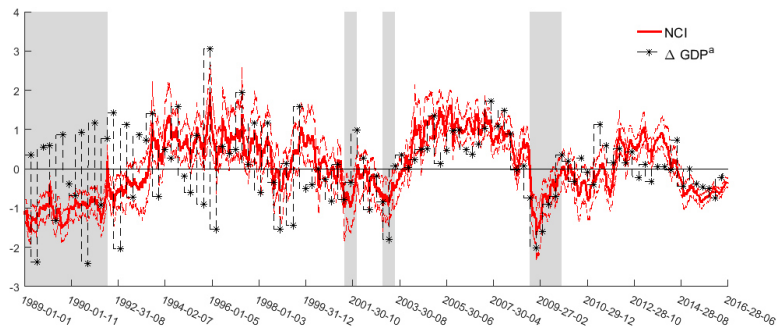
The Dynamic Factor Model

- ▶ y_t = quarterly GDP growth and daily newspaper topics
- ▶ Time aggregation using cumulator variable(s) (Harvey [1990] and Banbura et al. [2013])
- ▶ Estimation: Bayesian MCMC

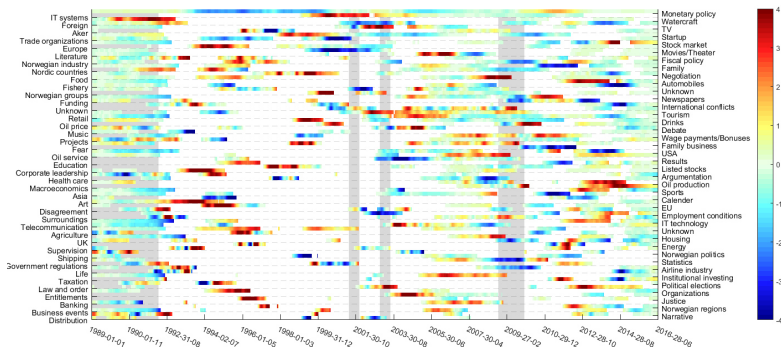
Two extensions

1. Time-varying parameters and threshold dynamics enforcing dynamic sparsity (Nakajima and West [2013])
 - ▶ Why: News topics might be informative about macro developments in some periods, but perhaps not always
2. Mixture Auxiliary Particle Filter
 - ▶ Why: For fast and efficient updating (nowcasting)

A Newsy Coincident Index (NCI)



News and index decompositions



Large degree of time-varying sparsity.
Topic contributions “tell” a plausible narrative

Summary 1

In-sample:

1. Almost perfect classification properties
2. News performs better than traditional economic indicators/models
3. Combined usage of news and time-varying parameters adds value
4. Provides high-frequency information about what drives/reflects economic fluctuations

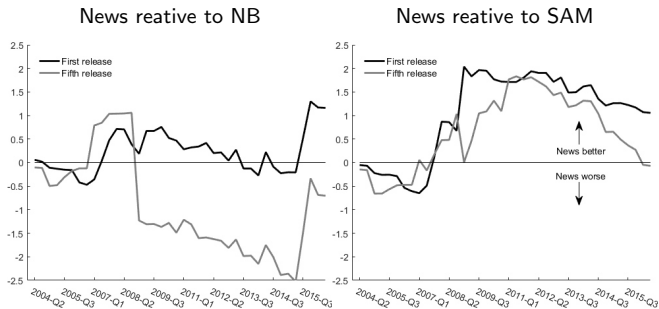
Nowcasting evaluation (out-of-sample)

RMSFE (using real-time data): Fifth release

Type	Nowcast for T_q				Backcast for T_q		
	$T_q - 2$				$T_q - 1$		
Quarterly information							
Daily information	$T - 80$	$T - 60$	$T - 40$	$T - 20$	T	$T + 10$	$T + 30$
RMSFE	0.537	0.538	0.547	0.520	0.497	0.502	0.500
Cum. imp. (%)		0.304	1.958	-3.153	-7.304	-6.472	-6.801
Relative RMSFE							
NCI^{notvp} / NCI	1.287**	1.324**	1.220	1.243**	1.107	1.105	1.104
NCI^{nosw} / NCI	1.083**	1.076***	1.092**	1.106**	1.074***	1.074***	1.077***
CI / NCI	1.113	1.106	1.072	1.241**	1.080	1.076	1.102
AR / NCI	1.142	1.139	1.121	1.491***	1.557**	1.543***	1.549***
RW / NCI	1.696***	1.692***	1.665***	1.553***	1.622***	1.608***	1.614***
SAM / NCI	1.020	0.974	0.934	1.008	1.000	0.991	0.996
NB / NCI					0.961	0.953	0.958

Nowcasting performance better than, or equally good as,
alternatives

News versus NB and SAM: Cumulative squared prediction error differences



Good around turning points?

Summary 2

Out-of-sample:

1. News superior to simple benchmarks, and equally good as sophisticated benchmarks and judgment
2. Combined usage of news and time-varying parameters adds value
3. Policy makers and the statistical office would have benefited from using news

Conclusion

1. The usage of newspaper data is novel and improves classification power and nowcasting
2. Enforcing sparsity and allowing for time-varying parameters in the DFM is important
3. News reduces noise

Part of a larger research agenda using text as data and news.
Similar news topics as used here:

- ▶ Predict cross section of daily stock returns, explain continuation patterns, and yield significant risk adjusted returns (Larsen and Thorsrud [2017])
- ▶ Explain quarterly long-run productivity developments (Larsen and Thorsrud [2015])

Application # 1

The Predictive Power of Google Searches in Forecasting US Unemployment

International Journal of Forecasting, October 2017

Literature

- ▶ The literature on US unemployment forecasting:
 - **linear** models: AR(I)MA-type models often augmented with some **leading indicators**
Montgomery, Zarnowitz, Tsay and Tiao (JASA, 1998)
 - **non-linear** models: STAR, SETAR, AAR, *(T)AR, unobserved non-linear component, etc.
Rothman (RESTAT, 1998), Proietti (CSDA, 2003) → “*testbed*”
 - **Okun's Law**: $\frac{\Delta Y}{Y} = k - c\Delta u$, Y =actual output, u =actual unemployment
 - more ‘state-of-the-art’ models: Barnichon and Nekarda (2012, BPEA) → use labor force flows
- ▶ In particular the typical **leading indicator** that so far seems to be the best for the US unemployment rate is the **Initial jobless Claims** (IC)

Our Contribution

- ▶ In this paper we suggest an *alternative leading indicator* to forecast the US unemployment rate
⇒ a **Google Index based on job-related web searches** on Google Search website
- ▶ Google data represent a “*measure*” of internet users’ **interest** in a particular phenomenon and can help us understand where the economy/society is going.
- ▶ We run an extensive out-of-sample forecasting *horse-race*, comparing the predictive power of linear forecasting models using:
 - ▶ Google Index (GI)
 - ▶ Initial Claims (IC)
 - ▶ Consumer Expectations (CE) and CE six months ahead (CE_t^{6m})
 - ▶ Employment Expectations in manufacturing (EEM) and non-manufacturing sector (EENM).
 - ▶ EPU index
- ▶ Forecast horizons: $h = 1, 2, \dots, 12$ months.

Unemployment rate (US and state level)

- ▶ **Monthly unemployment rate** (u_t) seasonally adjusted from BLS
 - ▶ Current Unemployment Statistics (national level)
 - ▶ Sample: **1948.1-2014.2** (794 obs.)
 - ▶ Local Area Unemployment Statistics (state level)
 - ▶ Sample: **1976.1-2014.2** (458 obs.)
- ▶ u_t for month t refers to:
 - ▶ people who **don't have a job**, but are **available for work**, in the week including the 12th of month t ...
 - ▶ ...and who **have looked for a job** in the previous 4 weeks (*reference week* included)

Google Trends data

- ▶ *Google Trends* (previously separated from *Google Insights for Search*) tracks relative changes in Google search queries from January 2004
- ▶ Google search queries
 - ▶ web searches
 - ▶ news searches
 - ▶ image searches
 - ▶ product searches
 - ▶ YouTube searches
- ▶ Different Geographical areas and levels (national, state and metropolitan area level) based on the originating IP address
- ▶ Available to the public for free download online at www.google.com/trends/

Features of *Google Trends* data

- ▶ Google Trends data represent **how many web searches** are done for a particular *keyword*, relative to the total # of searches in certain geographical area over time
- ▶ indicate the **likelihood of a random user to search for a particular keyword on Google** from a certain location at a certain time on a relative basis
- ▶ are gathered **using IP address information** from Google logs and **updated daily**
- ▶ are gathered only if the number of searches exceeds a certain **threshold of traffic**
- ▶ are such that **repeated queries** from a single user/IP over a short period of time are eliminated
- ▶ are **available world-wide** (by country, by region, by city)
- ▶ are **normalized** (divided by the total website traffic in the geographical area) $\Rightarrow$ comparability issues $\Rightarrow$ Search Volume Index (SVI)
- ▶ are **scaled** (from 0 to 100) dividing each data point by the maximum
- ▶ available both **weekly** and **daily** (only on last 90 days)

Aggregation, Normalization and Scaling

- ▶ For region r , the SVI for week τ is constructed aggregating the daily data for each day t . Given the search volume on a term “V”, $(V_{t,r})$ in region r on day t and the total search volume in that region $T_{t,r}$ we have the following for a total of T weeks
- ▶ Search Share for day t and week τ :

$$S_{t,r} = \frac{V_{t,r}}{T_{t,r}} \quad \text{and} \quad S_{\tau,r} = \frac{1}{7} \sum_{t=\text{Sunday}}^{\text{Saturday}} S_{t,r}$$

- ▶ Web Search Volume for week τ :

$$S_{\tau,r}^* = \frac{100}{\max_{\tau}(S_{\tau,r})} \frac{1}{7} \sum_{t=\text{Sunday}}^{\text{Saturday}} S_{t,r}$$

where $\tau = 1, \dots, T$

Matching options for job-related web queries in Google Trends

- ▶ typing `hospital jobs`: GI includes searches containing both `hospital` and `jobs` in any order and along with additional terms before (e.g. `Philadelphia hospital jobs`) and after (e.g. `hospital jobs in Denver`)
- ▶ typing `"San Diego jobs"`: GI includes searches with that specific order in quotes along with additional terms before and after (e.g. `"San Diego jobs" part time`)
- ▶ typing `"jobs in Oceanside" + "jobs in La Jolla"`: GI includes searches with either `"jobs in Oceanside"` or `"jobs in La Jolla"`, but not both
- ▶ typing `San Diego jobs - nanny`: GI includes searches with `San Diego jobs` in any order along with other terms before and after, but excluding those searches containing `nanny`

Which keyword(s) to forecast unemployment?

- ▶ We tried to imagine what an average American internet user hunting for a job would type in the Google bar. For example:
 - ▶ Tom is a high-school teacher and lives in PB (San Diego, CA). He has been fired a few weeks ago. He would probably look for “San Diego” teaching jobs
 - ▶ Hellen is a nurse and lives in Philadelphia (PA). She has been laid off three months ago and she would probably search for Philadelphia hospital jobs
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- ▶ What have in common all these web searches?

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- ▶ What have in common all these web searches?
- ▶ The keyword **“jobs”**

But what happened in October 2011?

But what happened in October 2011?

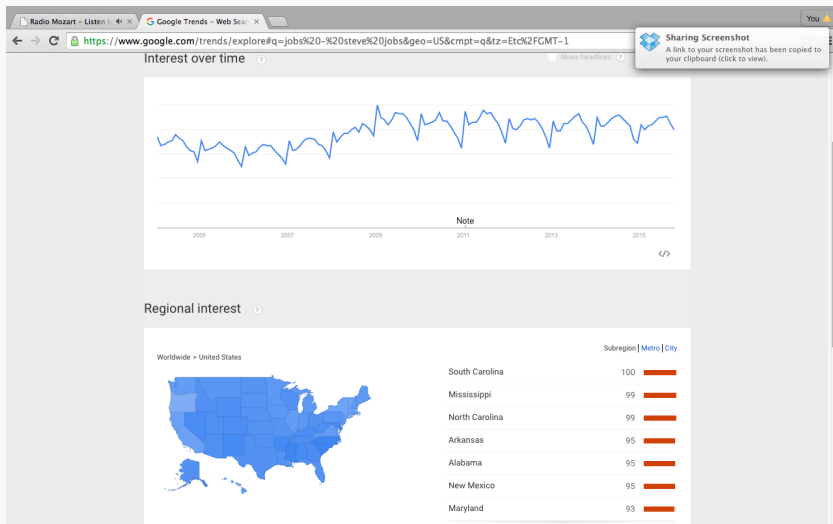


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- ▶ What have in common all these web searches?
- ▶ The keyword “jobs”
- ▶ Our preferred keyword is jobs - “Steve Jobs” (to avoid spike in Oct. 2011)

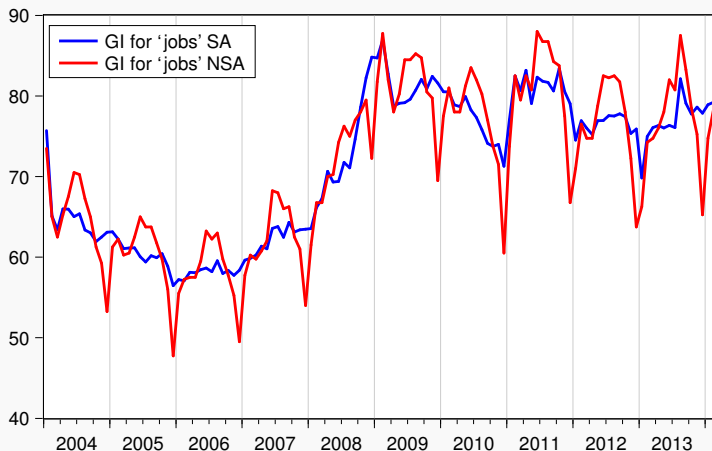
Web searches for keyword jobs - “Steve Jobs”

Weekly data: Jan. 2004 - Oct. 2015



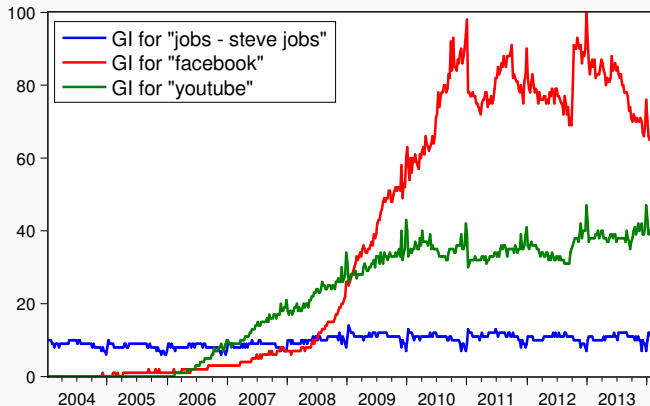
Peculiar seasonality

Monthly Google index for "jobs" - Sample: Jan. 2004 - Mar. 2014



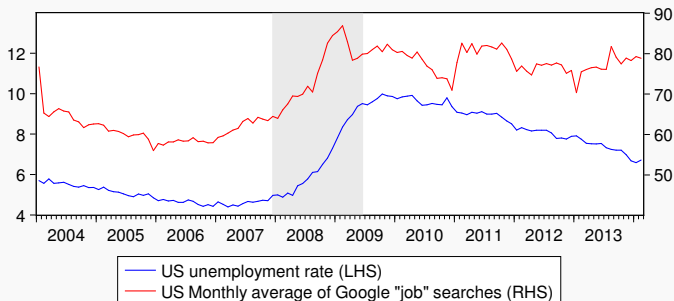
Relevance of our preferred keyword?

Incidence of keyword "jobs" vs other popular keywords (highest incidence)



Google Index for 'jobs' (US and state level)

- ▶ **Weekly Google Index (GI)** seasonally adjusted from Google Trends (available almost in real time)
⇒ suggested **Leading Indicator**
(Incidence of “*jobs*” queries on total web queries)
 - ▶ National level
 - ▶ Sample: **2004.1-2014.2** (122 obs.)
 - ▶ State level
 - ▶ Sample: **2004.1-2014.2** (122 obs.)



Data on Employment and Consumer Expectations

- ▶ Monthly data on Employment and Consumer **Expectations**
 - ▶ From Institute for Supply Management (ISM) Report on Business
 - ▶ Employment Expectations in manufacturing sector (EEM): Sample: **1967.1-2014.2** (566 obs.)
 - ▶ Employment Expectations in non-manufacturing sector (EENM): Sample: **1997.7-2014.2** (200 obs.)
 - ▶ From US Consumer Confidence survey of the Conference Board
 - ▶ Current Consumer Expectations (CE): Sample: **1990.1-2014.2** (290 obs.)
 - ▶ 6-month-ahead Consumer Expectations (CE6M): Sample: **1990.1-2014.2** (290 obs.)

The setup of the forecasting horse-race

- ▶ **Timing:** $T = R + P$ observations in **Survey time**
 - ▶ In the '**Long sample**' (1997.7-2014.2) we have $T = 200$
 - ▶ In the '**Short sample**' (2004.1-2014.2) we have $T = 122$
- ▶ The first R are used to estimate the models (**in-sample**) while the last P are used for **out-of-sample** evaluation.
- ▶ Want to predict u_t using linear AR models w/ and w/o exogenous leading indicators x_t :
 - ▶ $x_t = \{IC_t, \dots, IC_{t-k}\}$
 - ▶ $x_t = \{IC_{wj,t}, \dots, IC_{wj,t-k}\}, j = 1, \dots, 4, k = 1, 2$
 - ▶ $x_t = \{G_t, \dots, G_{t-k}\}$
 - ▶ $x_t = \{G_{wj,t}, \dots, G_{wj,t-k}\}, j = 1, \dots, 4, k = 1, 2$
 - ▶ $x_t = \{EEM_t, \dots, EEM_{t-k}\}$
 - ▶ $x_t = \{EENM_t, \dots, EENM_{t-k}\}$
 - ▶ $x_t = \{CE_t, \dots, CE_{t-k}\}$
 - ▶ $x_t = \{CE_t^{6M}, \dots, CE_{t-k}^{6M}\}$
 - ▶ $x_t = \{EPU_t, \dots, EPU_{t-k}\}$

The setup of the forecasting horse-race

- ▶ Forecasting scheme: we use a **rolling** scheme.
 - ▶ **'In Sample'** (2004.2-2007.2) $w/R = 37$
 - ▶ **'Out-Of-Sample'** (2007.3-2014.2) $w/P = 84$
- ▶ We use **direct** forecasts.
 - ▶ Benchmark $AR(p)$ with p selected by BIC recursively

$$y_{t+h}^h = \beta_0 + \beta_1(L)y_t + \eta_{t+h}, \quad t = 1, 2, \dots, T \quad (1)$$

- ▶ versus $AR-X(p)$ model w/ LI x_t with lags p and q selected by BIC recursively and sequentially ($p_{max} = q_{max} = 4$)

$$y_{t+h}^h = \beta_0 + \beta_1(L)y_t + \beta_2(L)x_t + \varepsilon_{t+h}, \quad t = 1, 2, \dots, T \quad (2)$$

- ▶ Also **Real-Time Forecasts** but only for the short sample: 2004.1-2014.2

OOS Results: Direct/Rolling scheme – Short sample

x_{it}	h=1	h=2	h=3	h=4	h=5	h=6	h=7	h=8	h=9	h=10	h=11	h=12
$AR(p)$	0.206	0.319	0.454	0.561	0.645	0.760	0.877	1.000	1.108	1.209	1.322	1.358
IC_t	0.847**	0.875**	0.836***	0.888**	0.941	0.975	0.990	1.012	1.002	0.992	0.982	1.024
$IC_{W1,t}$	0.891	0.932	0.918	0.946	0.962	0.969	0.984	1.000	1.013	1.014	1.023	1.023
$IC_{W2,t}$	0.884**	1.026	1.026	1.061	1.019	1.017	1.026	1.000	1.009	1.015	1.010	1.008
$IC_{W3,t}$	0.875**	0.912	0.899	0.972	0.983	1.010	1.026	1.014	1.059	1.063	1.077	1.097
$IC_{W4,t}$	0.835*	0.795*	0.781*	0.898	0.974	1.020	1.053	1.073	1.048	1.060	1.049	1.054
EEM_t	0.859*	0.887	0.877	0.932	1.007	1.041	1.075	1.061	1.068	1.101	1.088	1.142
$EENM_t$	0.824**	0.777**	0.746**	0.751***	0.797**	0.818**	0.834*	0.848	0.845	0.842	0.831	0.867
CE_t	0.889	0.858	0.771	0.765	0.755	0.756	0.742	0.719	0.712	0.718	0.710	0.700
CE_t^{6M}	0.894	0.873	0.805	0.811	0.826	0.801	0.760	0.734	0.722	0.719	0.706	0.754
$G1_t$	0.821*	0.765*	0.656*	0.652*	0.644*	0.662*	0.659*	0.665*	0.680*	0.678*	0.673*	0.685*
$G1_{W1,t}$	0.952	0.883	0.967	1.066	1.195	1.276	1.311	1.287	1.252	1.225	1.204	1.231
$G1_{W2,t}$	0.901	0.975	0.967	1.006	1.086	1.051	1.021	1.008	0.971	0.989	0.980	0.907
$G1_{W3,t}$	0.866*	0.802*	0.682*	0.687*	0.704*	0.710*	0.700*	0.682*	0.683*	0.709*	0.731*	0.766*
$G1_{W4,t}$	0.843**	0.777*	0.672*	0.694*	0.691*	0.716**	0.798**	0.798**	0.813*	0.798*	0.793*	0.808*

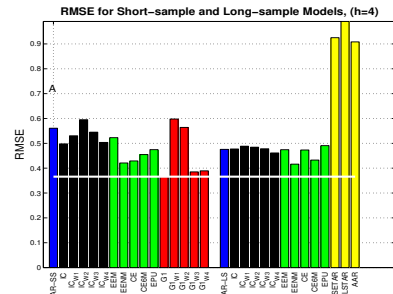
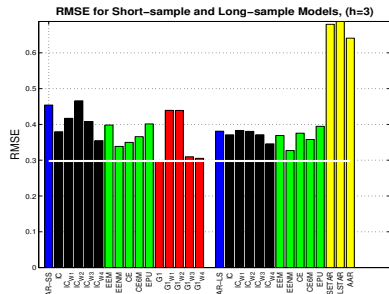
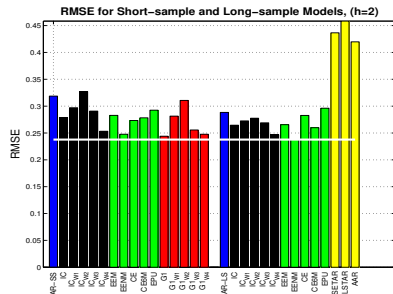
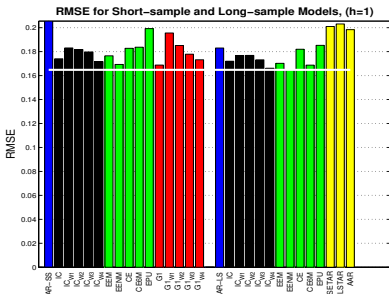
RMSFE in red.

OOS Results: Direct/Rolling scheme – Long sample

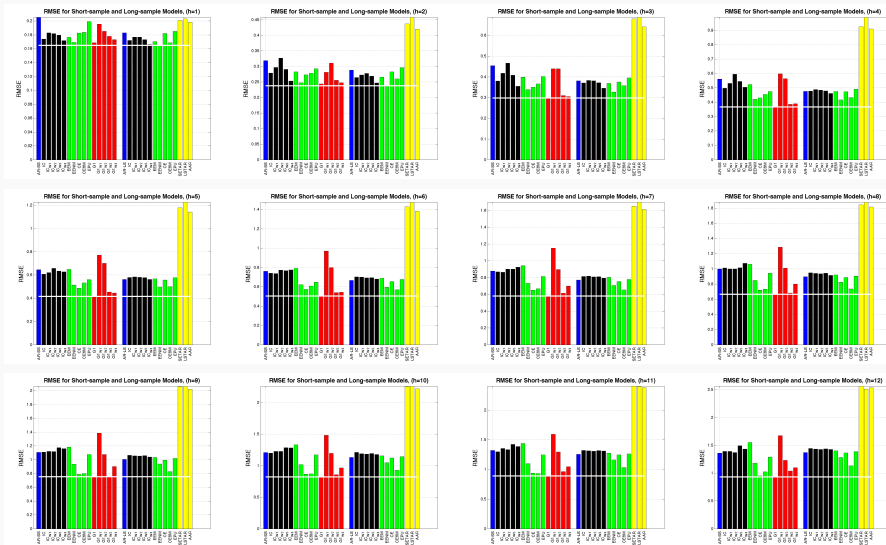
x_{it}	h=1	h=2	h=3	h=4	h=5	h=6	h=7	h=8	h=9	h=10	h=11	h=12
$AR(p)^*$	0.206	0.319	0.454	0.561	0.645	0.760	0.877	1.000	1.108	1.209	1.322	1.358
$AR(p)$	0.891**	0.905*	0.839*	0.848*	0.871*	0.877**	0.880**	0.896**	0.909*	0.936*	0.949*	1.009 [†]
IC_t	0.837*	0.830	0.817	0.851	0.898	0.925	0.927	0.947	0.960	1.001	1.000	1.061
$IC_{W1,t}$	0.860	0.855	0.844	0.871	0.906	0.922	0.931	0.940	0.953	0.984	0.995	1.055
$IC_{W2,t}$	0.861	0.871	0.838	0.864	0.901	0.912	0.921	0.935	0.951	0.980	0.990	1.049
$IC_{W3,t}$	0.843*	0.844	0.817	0.853	0.896	0.914	0.924	0.942	0.956	0.987	0.998	1.057
$IC_{W4,t}$	0.809**	0.774*	0.761	0.823	0.872	0.895	0.904	0.916	0.937	0.973	0.992	1.051
EEM_t	0.828**	0.833	0.812	0.846	0.879	0.906	0.916	0.920	0.934	0.957	0.965	1.031
$EENM_t$	0.802**	0.746*	0.720	0.742	0.770	0.784	0.806	0.824	0.846	0.866	0.877	0.943
CE_t	0.886	0.887	0.827	0.844	0.864	0.862	0.859	0.885	0.897	0.928	0.942	1.003
CE_t^{6M}	0.821**	0.815	0.788	0.772	0.776	0.750	0.743	0.736	0.748	0.768	0.782	0.834
$SETAR(2)$	0.978	1.370 [†]	1.497	1.650	1.827 [†]	1.879 [†]	1.881 [†]	1.842 [†]	1.858 [†]	1.865 [†]	1.818 ^{††}	1.886 ^{††}
$LSTAR(2)$	0.988	1.438 ^{††}	1.515 [†]	1.767 [†]	1.901 [†]	1.942 [†]	1.935 [†]	1.878 [†]	1.864 ^{††}	1.869 ^{††}	1.814 ^{††}	1.850 ^{††}
$AAR(2)$	0.965	1.317 [†]	1.412	1.620	1.770	1.818	1.839 [†]	1.815 [†]	1.824 [†]	1.835 [†]	1.799 [†]	1.866 [†]

* $AR(p)$ estimated over the short sample

Out-of-sample Results with $G1$ (Direct/Rolling)

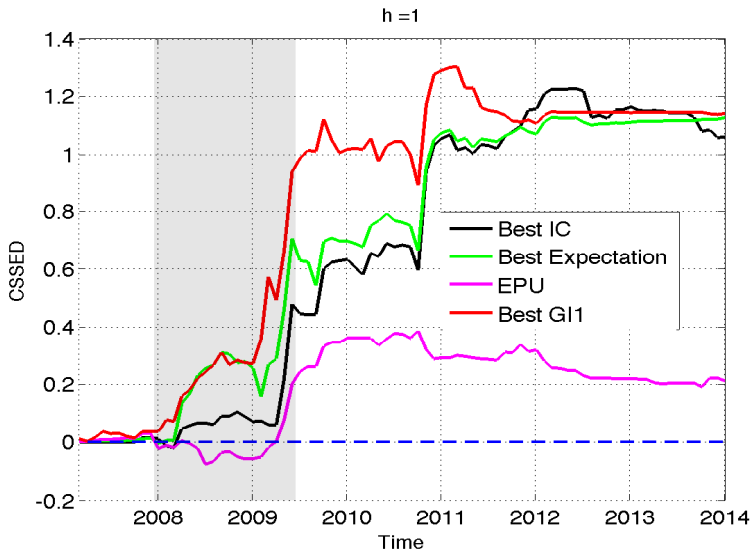


Out-of-sample Results with $G1$ (Direct/Rolling)



Predictive Relative Performance with CSSED

$CSSED_{m,\tau} = \sum_{\tau=R}^T (\hat{e}_{bm,\tau}^2 - \hat{e}_{m,\tau}^2)$ 'best' model w.r.t. the $AR(P)$ benchmark



Robustness checks

- 1) Exercise with Real Time Data
- 2) Comparison with Survey of Professional Forecasters
- 3) Alternative keywords to “jobs” (G1): “job center” (G2), and combination of “jobs” + “collect unemployment” + “job center” (G4).
- 4) Falsification test: False/Fake/Placebo GI “dos” (G3), highest in-sample correlation with u_t (Google Trends Correlate) but completely unrelated to job searches (‘Dept. of State’ or ‘Disk Operating System’)
- 5) Non-linear models
- 6) Robustness over GI across different IP addresses and different days of download: 12 days from 2 IPs. 24 GI, plus 3 averages: GI_A (overall), GI_A1 (over IP1), GI_A2 (over IP2)
- 7) State-level forecasts

Comparison with the SPF

Sample: 2007:Q2-2014:Q1

- ▶ We also compared our forecasting models with the Survey of Professional Forecasters (SPF) (mean and median)
- ▶ At the 'middle' of $Q(t)$ (around Feb, May, Aug and Nov 15) SPF issues forecasts for $Q(t+1)$ to $Q(J+5)$
- ▶ We compare SPF^{median} , and SPF^{mean} with different Quarterly forecasts of the US unemployment built as **simple averages** of the monthly forecasts of our best models:
 - ▶ Best model using Google Index
 - ▶ Best model without Google using Initial Claims
 - ▶ Short sample
 - ▶ Long sample
 - ▶ Best model without Google using Expectations
 - ▶ Short sample
 - ▶ Long sample
 - ▶ State-of-the-art model using flows, i.e. SSUR2 (2-state Steady-state UR model) by Barnichon and Nekarda (2012)

Comparison with the SPF (Cont.)

Sample: 2007:Q2-2014:Q1

- For each model (except for the SSUR2) we compute the quarterly forecast as the average of the realized unemployment rate for the first month of the quarter and the 1-month- and 2-month-ahead forecasts generated in the middle of the second month of the quarter
 1. Around the 12th of the second month of $Q(t)$, e.g. 2007.5: forecast 1- and 2-month ahead

$$[u_t + \hat{u}_{t+1|t} + \hat{u}_{t+2|t}]/3 \Rightarrow \mathbf{UR}_{Q(t)}^{\text{QTR}}$$

is our forecast for $Q(t)$

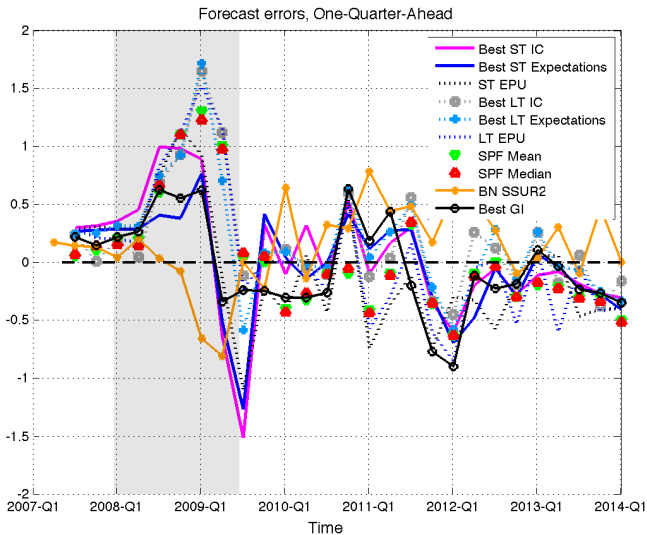
2. For longer horizon, i.e. $Q(t+j)$ with $j \geq 1$, $\mathbf{UR}_{Q(t+j)}^{\text{QTR}}$ is average of 3- to 5-month-ahead monthly forecasts ($Q(t+1)$), 6- to 8-month-ahead forecasts ($Q(t+2)$) and 9- to 11-month-ahead forecasts ($Q(t+3)$).

Comparison with the SPF - Sample: 2007:Q2-2014:Q1

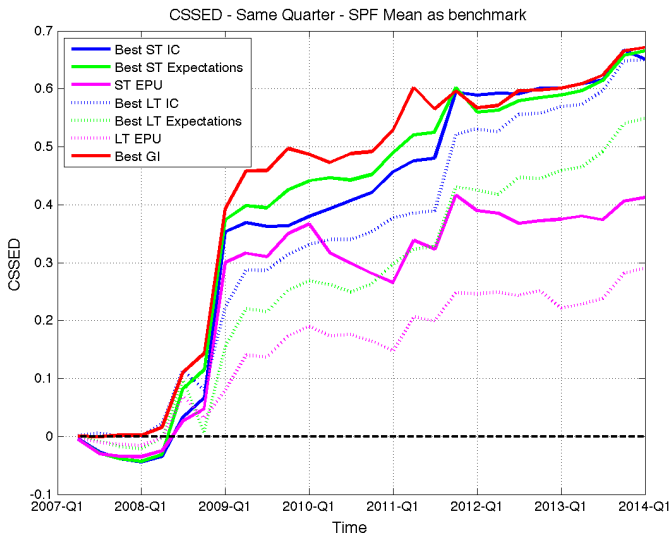
	$\hat{Q}_{t+0}$		$\hat{Q}_{t+1}$		$\hat{Q}_{t+2}$		$\hat{Q}_{t+3}$	
x_{it}	RMSFE	Rank	RMSFE	Rank	RMSFE	Rank	RMSFE	Rank
IC_t	0.149	6	0.524	9	0.984*	22	1.269*	18
$IC_{W1,t}$	0.164	14	0.566	18	0.911	17	1.286*	21
$IC_{W2,t}$	0.168	18	0.617	24	0.972	21	1.285*	20
$IC_{W3,t}$	0.151	7	0.573*	19	1.035*	24	1.431*	23
$IC_{W4,t}$	0.141	3	0.548	14	1.032*	23	1.433*	24
EEM_t	0.142	4	0.536*	12	0.938	20	1.303	22
$EENM_t$	0.139	2	0.422	4	0.711	5	1.004	6
CE_t	0.181**	22	0.5	7	0.759*	9	0.99	5
CE_t^{6M}	0.19*	24	0.533	11	0.732	6	0.944	4
$G1_t$	0.138	1	0.404	3	0.598	1	0.781	1
$G1_{W1,t}$	0.162	13	0.739	25	1.348	25	1.717	25
$G1_{W2,t}$	0.188**	23	0.608	23	0.886	14	1.221	13
$G1_{W3,t}$	0.147	5	0.393	2	0.639	2	0.881*	3
$G1_{W4,t}$	0.155*	10	0.439	5	0.656	3	0.87	2
$SSUR2$	0.152	8	0.377	1	0.671	4	1.008	7
SPF^{med}	0.199**	25	0.474	6	0.754	8	1.109	9
$IC_t - LS$	0.166	15	0.575	20	0.907*	16	1.268*	17
$IC_{W1,t} - LS$	0.178	21	0.588	22	0.916	19	1.26*	16
$IC_{W2,t} - LS$	0.172	19	0.582	21	0.913	18	1.27*	19
$IC_{W3,t} - LS$	0.159	12	0.555*	16	0.894*	15	1.255*	15
$IC_{W4,t} - LS$	0.158	11	0.553	15	0.882*	13	1.24*	14
$EEM_t - LS$	0.167	17	0.555*	17	0.861	12	1.215	12
$EENM_t - LS$	0.153	9	0.507	8	0.785	10	1.129	10
$CE_t - LS$	0.178**	20	0.548	13	0.821*	11	1.149	11
$CE_t^{6M} - LS$	0.166*	16	0.528	10	0.737	7	1.025	8

Comparison with the SPF - Quarter ($Q(t+1)$)

Forecast errors ($\hat{e}_{k,t+\tau} = u_{t+\tau} - \hat{u}_{k,t+\tau}|t$) of 'best' models-2007:Q2-2011:Q2

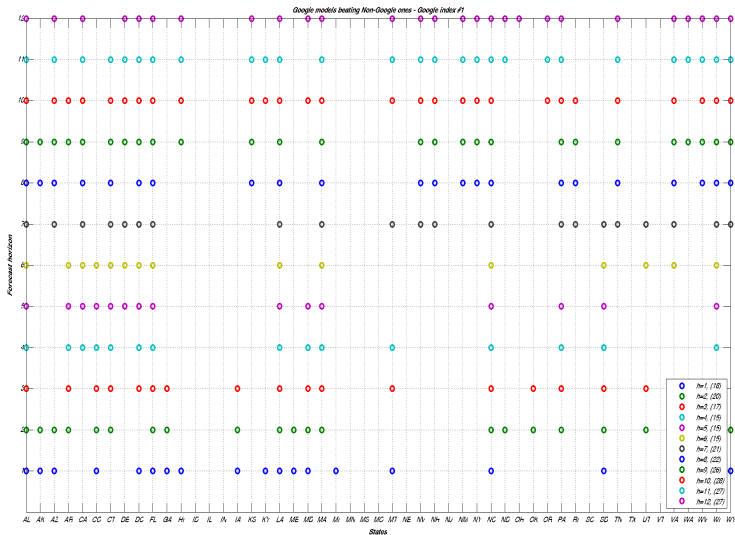


Comparison with the SPF - CSSED vs Mean SPF



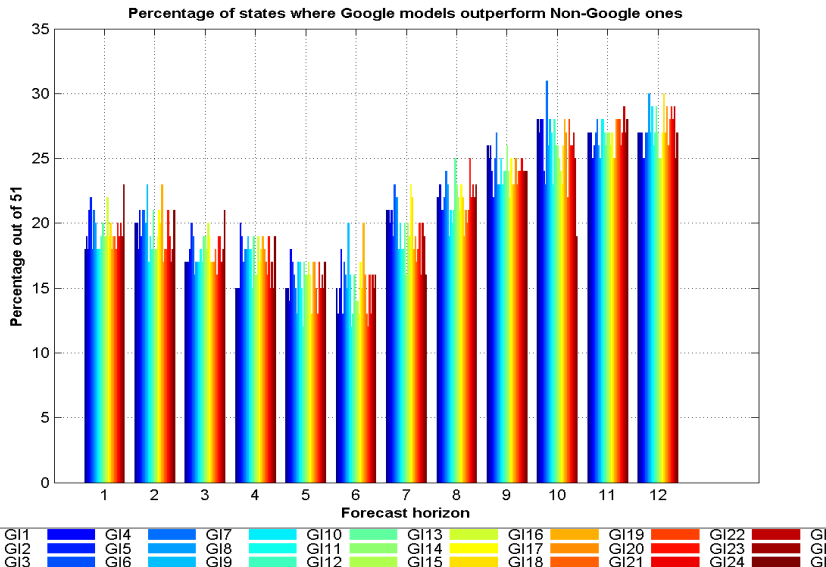
Robustness across states

Number of Google models beating Non-Google ones by state



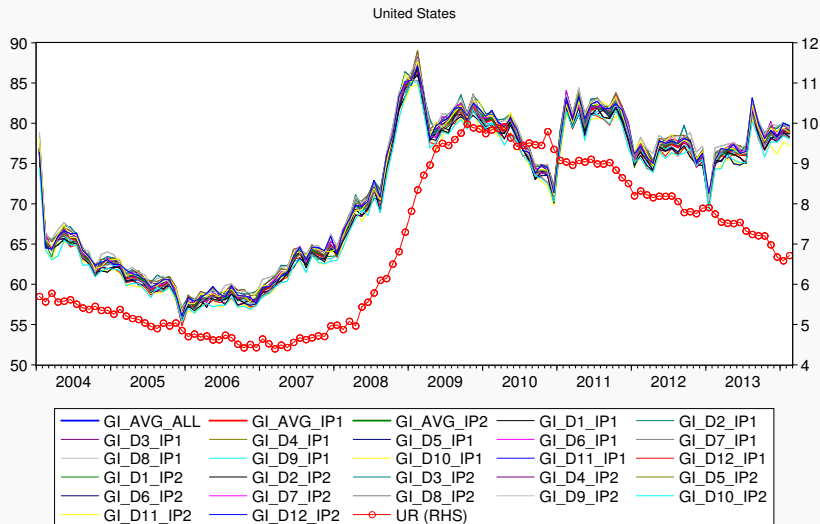
Robustness across states and GIs

% of Google models outperforming IC models across GI & IPs & days of download



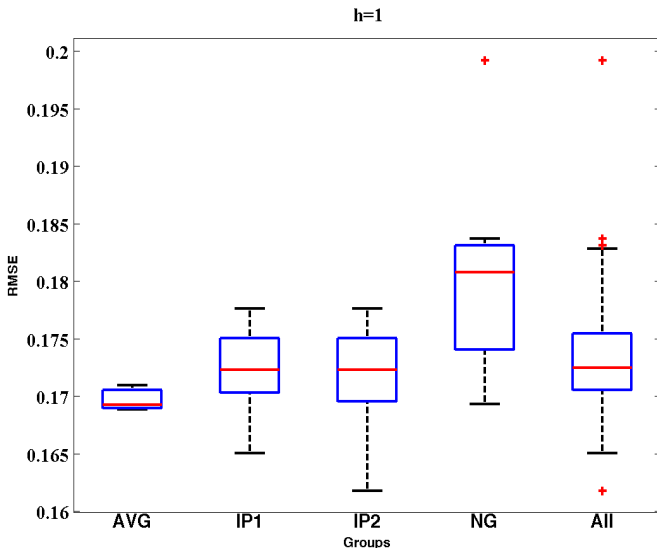
Robustness across GIs

Google indexes for the US across IPs and days of download



Robustness across GIs (Cont.)

Relative performance across GI downloaded in different days/IPs



Text mining with Twitter: inflation expectations

Can We Measure Inflation Expectations Using Twitter?

*By C. Angelico, M. Miccoli, J. Marcucci, and F. Quarta
Bank of Italy, mimeo, 2019*

Motivation

Inflation Expectations play a crucial role in macroeconomics:

- ▶ Key to understand consumption and investment choices
- ▶ Informative on the effectiveness of central bank actions (both at short and long term)

Available sources of expectations:

- ▶ **Survey-based**: “true” expectations, but low frequency
- ▶ **Market-based**: high frequency, but risk premia

Can we use social media to elicit inflation expectations?

Could combine ‘true’ expectations & ‘high frequency’

Data selection – Relevant Keywords

Keywords (EN/IT) to select tweets that talk about inflation/deflation

- ▶ *price(s), cost of living*
 - ▶ *prezzo, prezzi, costo della vita*
- ▶ *expensive bills, inflation, expensive, high prices, high-prices, high gas prices, higher bill, higher rents, high gasoline price, high oil prices, high gas bills*
 - ▶ *caro bollette, inflazione, caro, caro prezzi, caroprezzi, benzina alle stelle, bolletta salata, caro affitti, caro benzina, caro carburante, caro gas*
- ▶ *deflation, disinflation, sale(s), less expensive, less expensive bills*
 - ▶ *deflazione, disinflazione, ribassi, ribasso, meno caro, bollette più leggere*

Data

- ▶ 7.4 millions of tweets
- ▶ Sample: 1 June 2013 – 25 June 2018
- ▶ About 780,000 individual users
- ▶ Tweets contain one (or more) of the keywords
- ▶ Full text and the meta-data (i.e. users' bio)

Intuition

- ▶ Tweets reflect info on current or future prices
- ▶ They can be inputs to the expectations formation process

Twitter-based inflation expectations

Aim

- ▶ Need to select tweets related to price dynamics
- ▶ Build meaningful indexes of inflation expectations

Many tweets $\Rightarrow$ “noise” (i.e. advertisements)

Two-step procedure

1. Topic analysis

- ▶ Exploit the full text of the tweets
- ▶ Isolate valuable signals and filter the tweets of interest

2. Dictionary-based approach

- ▶ Exploit the semantic of the keywords
- ▶ Build a set of indicators on the filtered data

Topic analysis: Details

Steps:

1. Text cleaning

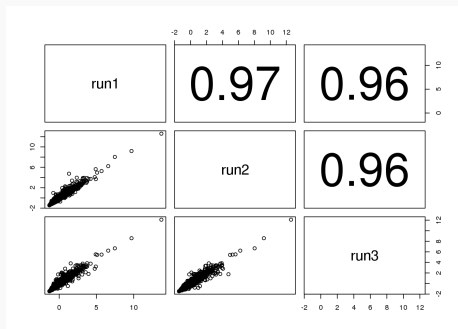
- ▶ Stopwords, punctuation, stemming, IDF filtering

2. User-pooling

- ▶ To get a set of documents given the shortness of tweets

3. Three runs of the LDA

- ▶ Independent runs to deal with the stability of the topics



Topics discovered by LDA: valuable signals?

Smartphone e-commerce		Topic 1		Topic 2	
Italian	[English]	Italian	[English]	Italian	[English]
iphon		inflazion	[inflat]	salar	[wage]
apple		ribass	[sale]	inflazion	[inflat]
caratterist	[feaut]	petrol	[oil]	deflazion	[deflat]
galaxy		bors	[stock exchange]	lavor	[job]
samsung		ital		claud	
ital		deflazion	[deflat]	aument	[increas]
nuov	[new]	istat		bass	[low]
galaxy		cresc	[growth]	ital	
smartphone		aument	[increas]	disoccupaz	[unempl]
offert	[offer]	merc	[market]	econom	[economy]

- ▶ 150 topics (but we tried many)
- ▶ Focus on tweets assigned to these 2 topics (inflation/deflation)
- ▶ Final dataset: 678,476 tweets

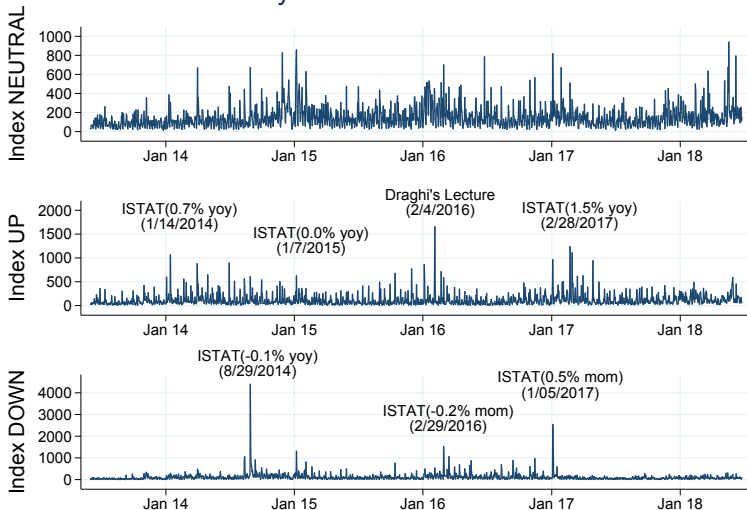
Second step: Dictionary-based approach

How can we aggregate tweets to get insights on agents' beliefs?

- ▶ Keywords reflect a message on the direction of price changes
- ▶ Index computed as daily raw count of tweets
 - ▶ **Index Neutral:** *price(s) + cost of living*
 - ▶ **Index Up:** *inflation + expensive bills + expensive + high prices + high gas prices + higher bill + higher rents + high gasoline prices + high gas bills*
 - ▶ **Index Down:** *deflation + disinflation + sale(s) + less expensive + less expensive bills*

Dictionary-based Indexes

Dictionary-based Inflation Indexes



Twitter-based directional indicators

$$\pi_0^e = \text{Index Up} - \text{Index Down}$$

1. Infl. Exp. #1

- ▶ filtering on event dummies, standardization, winsorizing, backward looking MA 10, 30, 60 days

2. Infl. Exp. #2

- ▶ standardization, winsorizing, backward looking MA 10, 30, 60 days

3. Infl. Exp. #3

- ▶ exponential smoothing

$$\pi_{ln}^e = (\ln(\text{Index Up} + 1) - \ln(\text{Index Down} + 1))$$

4 Infl. Exp. #4

- ▶ backward looking MA 10, 30, 60 days

Twitter-based directional indicators



Do we capture inflation expectations?

Compare Twitter-based indicators with:

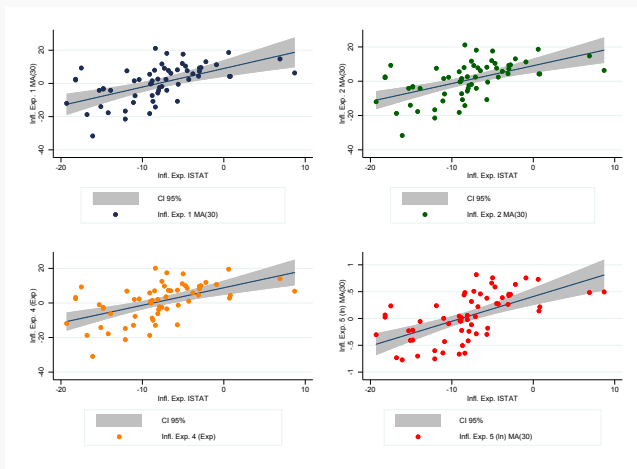
- ▶ **Survey-based**

- ▶ ISTAT monthly consumers' expectations
- ▶ Qualitative expectations over next 12m inflation
- ▶ First best, but short time series

- ▶ **Market-based**

- ▶ Daily inflation swap rates with 1 year maturity
- ▶ Quantitative expectations
- ▶ Second best, given the caveat of risk premia

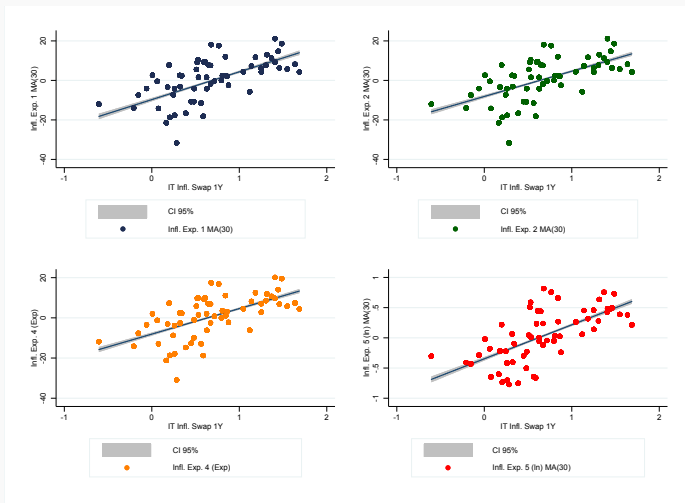
Twitter vs survey-based inflation expectations



Correlations: Twitter and ISTAT Infl. Exp

Infl. Exp. 1 MA(30)	Infl. Exp. 2 MA(30)	Infl. Exp. 3 (Exp -0.1)	Infl. Exp. 4 (ln) MA(30)
0.512***	0.513***	0.418***	0.501***

Twitter vs market-based inflation expectations



Correlations: Twitter Infl. Exp. and IT Infl. Swap 1Y

Infl. Exp. 1 MA(30)	Infl. Exp. 2 MA(30)	Infl. Exp. 3 (Exp-0.1)	Infl. Exp. 4 (ln) MA(30)
0.546***	0.539***	0.535***	0.625***

Informativeness exercises

	(1)	(2)	(3)	(4)	(5)
Lagged $E_t^{IS\text{STAT}} \pi_{t,t+12}$	0.596*** (0.0425)	0.309*** (0.0664)	0.437*** (0.0556)	0.294*** (0.0540)	0.265*** (0.0543)
Inflation swap 1y rate, MA(10)		5.765*** (1.043)		4.446*** (1.093)	5.046*** (0.893)
Infl. Exp. 4 (ln), MA(10)			4.493*** (0.954)	2.301*** (0.688)	2.033*** (0.704)
Flash infl. estimate					-355.7** (159.2)
Constant	-3.338*** (0.819)	-9.571*** (1.298)	-4.590*** (0.453)	-8.787*** (1.148)	-9.223*** (1.083)
Observations	61	61	61	61	61
Adj. R^2	0.354	0.532	0.476	0.550	0.562
Root MSE	4.566	3.885	4.109	3.812	3.76

- ▶ IS_t^{1y} and $Infl.exp._t$ have explanatory power and correct sign
- ▶ Twitter index provides small, but additional, information

Conclusion

Many other papers deal with text mining and economics

- ▶ Gentzkow, Kelly and Taddy (2017), 'Text as Data', NBER wp 23726
- ▶ Larsen, Thorsrud, and Zhulanova, (2019), 'News-driven inflation expectations and information rigidities'
- ▶ Shapiro, Sudhof, and Wilson, (2018), 'Measuring News Sentiment', Federal Reserve Bank of San Francisco, working paper 2017-01.
- ▶ Larsen and Thorsrud, (2019), 'Business cycle narratives', CESIFO working paper 7468
- ▶ Lima, Godeiro, and Mohsin, (2019), 'Time-varying Dictionary and the predictive power of FED minutes', working paper, University of Tennessee.
- ▶ Scott and Madeley, (2010), 'Text Mining and the information content of Bank of Canada Communications', Bank of Canada working paper 2010-31.
- ▶ Goshima, Ishijima, Shintani, and Yamamoto, (2018), 'Forecasting Japanese Inflation with news-based business cycle indexes', working paper.
- ▶ Bollen, Mao, and Zeng, (2011), 'Twitter mood predicts the stock market', Journal of Computational Science, 2, 1-8.
- ▶ Sabouni (2018), 'The Rhythm of Markets', Claremont Graduate University working paper
- ▶ ...

Conclusion

Key messages to take home

- ▶ Big Data and text corpora have **lots of potential** for official statistics, nowcasting and forecasting
- ▶ There are many opportunities for both public institutions and private companies
- ▶ We have just seen some applications and very simple models.
- ▶ There are already more complicated models: for example Dynamic Topic Models (DTM) where topics can evolve over time
- ▶ It will always be a **signal to noise** problem!

Conclusion (Cont.)

Key messages to take home

- ▶ Wherever you will end up working please remember that collaboration and cooperation always bring to a better equilibrium (in a game theoretic sense).
- ▶ **Collaboration** among institutions (NSIs, CBs, Private Companies, etc.) and among (data) scientists with different skills
- ▶ Collaboration at the European (or global) level
- ▶ **Partnerships** between different stakeholders (government, academia, private sector, central banks, ...)

Let's get ready to dive into this ocean of Big Data!

The search for insights... in an ocean of data.

